

SOEN 220 – Network Structure & Topologies

Data Communication & Networks | Clean Revision Notes

1. Intranet and Extranet

Intranet

A private network that is accessed **only by members of a firm / organisation**. It uses the same technologies as the Internet (TCP/IP, web browsers, etc.) but is restricted to internal users. Typical uses: company policies, internal email, shared files, HR systems.

Extranet

A private network that is accessed by members of the organisation **and by trusted external partners** (suppliers, customers, contractors). It extends the intranet outward in a controlled way. Access is usually protected by passwords, VPNs or firewalls.

Simple memory aid: **Intra** = inside only. **Extra** = inside + selected outsiders.

2. Network Topology

Topology

The virtual shape or structure of a network — how devices are arranged and connected.

Physical topology

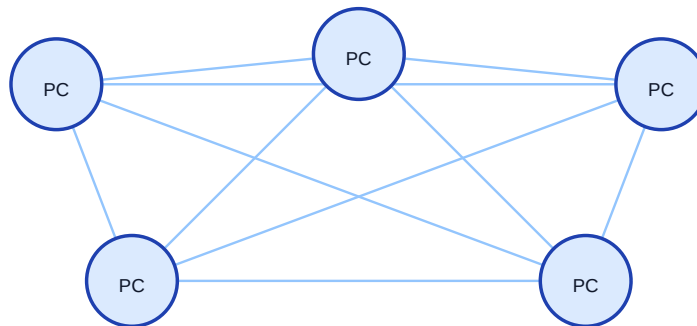
The actual way in which a network is laid out physically (the real cables and device positions).

When designing a network, three important factors are always considered:

- **Performance** – how fast and efficiently the network responds and transfers data
- **Reliability** – how correctly and consistently the network works, and how well it recovers from failures
- **Security** – the ability to protect data and resources from unauthorised access or attacks

3. Mesh Topology

In a mesh topology every device has a dedicated point-to-point link to every other device. A dedicated link carries traffic only between the two devices it connects.



Every device linked to every other (full mesh)

Full mesh – each node connected to all others

Advantages

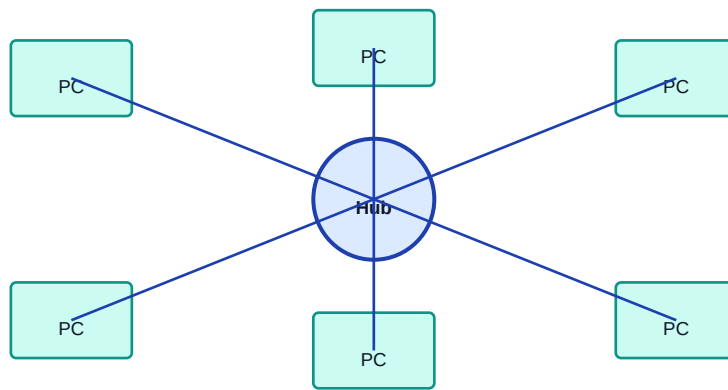
- Use of dedicated links guarantees that each connection carries only its own data (no sharing of bandwidth on that link)
- Robust – a fault in one link does not incapacitate the entire system
- Privacy and security are high because data travels on a private path between two devices

Disadvantages

- Requires a lot of cabling and many I/O ports
- Installation and reconnection are difficult
- Hardware required is expensive

4. Star Topology

Each device has a dedicated point-to-point link only to a central controller (usually a hub or switch). Devices do not connect directly to each other; all traffic goes through the centre.



Star – all devices linked to a central hub/switch

Advantages

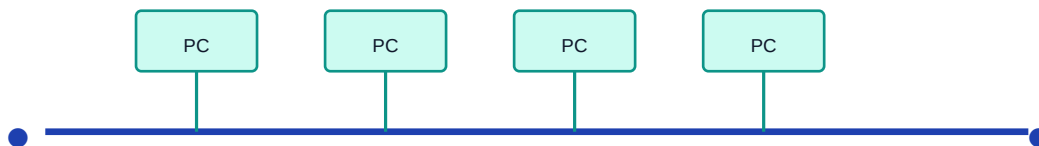
- Less expensive than mesh (each device needs only one link)
- Easy to install and reconfigure – add or remove a device without disturbing the rest
- Robustness – failure of one link or one device does not affect the others
- Easy to identify and isolate faults

Disadvantages

- Dependency on the central hub – if the hub fails, the whole network fails (single point of failure)

5. Bus Topology

All devices share one long cable (the backbone). Signals travel along the cable in both directions. A multipoint configuration: one cable, many stations connected to it.



Shared backbone cable

Bus – one shared backbone cable with stations attached

Advantages

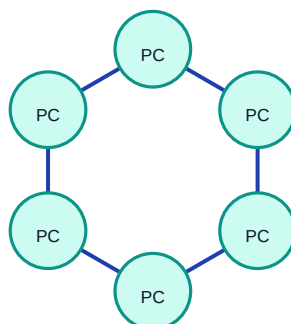
- Easy to install
- Uses less cabling than star or mesh → cheaper for small networks

Disadvantages

- Difficult reconnection and fault isolation
- A break in the bus cable can stop all transmission
- Signal reflection at the ends / taps can cause degradation of quality

6. Ring Topology

Each device has a dedicated point-to-point connection with only the two devices on either side of it. Data travels in one direction around the ring (or sometimes in both directions in dual rings).



Ring – devices form a closed loop

Repeaters are often used to regenerate the signal as it travels around the ring so that it does not become too weak.

Advantages

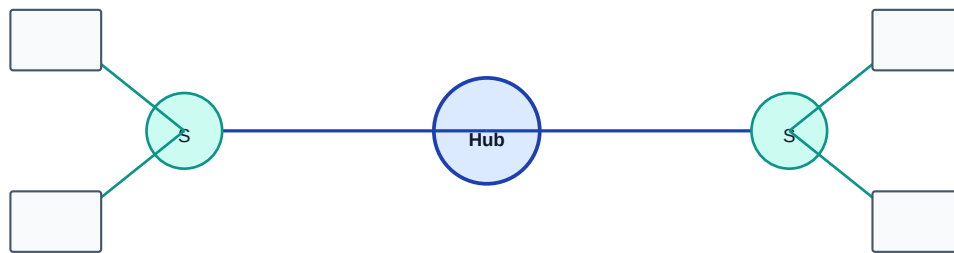
- Easy to install and reconfigure
- Fault isolation can be simplified (a break can be detected by the neighbouring stations)

Disadvantages

- A break in one point can disrupt the whole ring (unless a dual-ring or bypass mechanism is used)

7. Hybrid Topology

A hybrid topology combines two or more different types of topology. A common example is a star-of-stars: several star networks connected together through a central hub or switch.



Hybrid = combination of two or more topologies (e.g. star + star)

Hybrid example – multiple stars joined together

Hybrid designs are used when no single topology is ideal for the whole organisation. Different departments can use the topology that suits them best, then be linked together.

8. Factors to Consider When Choosing a Topology

1. **Nature of the network** – size, purpose, number of users
2. **Cost** – budget for cables, devices and installation
3. **Length of the cable needed** – longer runs cost more and can affect signal quality
4. **Future growth** – will it be easy to add more devices later?
5. **Cable type** – twisted pair, coaxial, fibre, etc., and the distances they support

9. Network and Internet (quick reminder)

A **network** is the connection of computers and devices for the purpose of communication and sharing of resources. The **Internet** is the global interconnection of networks — a network of networks that spans the world.

10. Quick Revision Checklist

- Intranet = private, organisation only. Extranet = organisation + trusted outsiders.
- Topology = the shape/structure of the network (physical or logical).
- Mesh = every device linked to every other → very reliable, very expensive.
- Star = all devices linked to a central hub → easy to manage, hub is a single point of failure.
- Bus = one shared cable → cheap, but a cable break can stop everything.
- Ring = closed loop → simple, but one break can affect the ring.
- Hybrid = mix of topologies → flexible for large organisations.
- Always weigh performance, reliability, security, cost and future growth when choosing.

These notes expand the class material on network topology and structure for SOEN 220. Draw the five topology diagrams yourself a few times — that is the fastest way to lock them in memory before the CAT.